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Printed Newton's egg curved monopole antenna for ultrawideband applications

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Abstract: The edge curve of a planar antenna significantly influences its radiation properties. In this study, a novel egg curve-based design of printed monopole antenna of size 40×40 mm^2 is proposed for ultrawideband (UWB) applications. The egg shape of the proposed antennas is obtained by Newton's diverging parabolas curve referred in this study as Newton's egg curve. The design equations of the proposed antenna are developed. The variation in impedance bandwidth (BW) with various parametric variations of design curve is studied. The proposed Newton's egg curved monopole antenna (NECMA) is designed, fabricated and measured to validate the theoretical design. Good agreement between the simulated results and the measured ones is observed. An empirical formula is also proposed to approximately determine the frequency corresponding to the lower edge of −10 dB operating BW. The results show that the proposed NECMA can obtain the measured BW of 116.55% (2.9–11.0 GHz) along with stable gain of about 2.2–2.9 dBi. The proposed antenna exhibits almost omnidirectional radiation patterns with low cross polarisation in all directions and consistent group delay over the entire UWB bandwidth (3.1–10.6 GHz).

1 Introduction

Since the US Federal Communications Commission (FCC) released the use of frequencies from 3.1 to 10.6 GHz for ultrawideband (UWB) communications in 2002 [1], a great deal of attention has been focused on the components needed for such systems. Among them, broadband antennas are required for emerging applications using the UWB frequency range. To serve such applications in wireless communications and medical imaging systems, monopole antennas have been widely used and are the most attractive candidates due to stable radiation patterns, low profiles, low manufacturing costs, lightweight, simple structure and ease to fabricate. However, they generally have a relatively narrow bandwidth.

Many techniques to improve the impedance bandwidth have been studied, such as parasitic elements [2], bevelling techniques [3] and the use of two or three feeds [4]. However, the antennas with parasitic elements suggested previously [2] are too large to be used in most practical applications. The probe-fed antennas in [3] occupy a large volume because their ground planes are orthogonal to the radiator. In [5], an inverted T-shaped slot in printed UWB monopole is used along with an inverted T-shaped conductor-backed plane to excite additional resonance through which the bandwidth of 130% is achieved. The coplanar waveguide (CPW)-fed E-shaped antenna [6] is not suitable for integration with a circuit board as it uses a large ground plane. A koch fractal slot in the ground plane for microstrip line fed antenna is proposed in [7] for bandwidth enhancement (2.33–6.19 GHz).

Thus, broadband printed monopole antennas have attracted many researchers. Therefore, a large number of planar antenna designs such as square [4], circular [8], elliptical [9], triangular [10] and cone shape [11] have been investigated for UWB communications. In [12], a planar elliptical dipole antenna is proposed but its gain dropped continuously over UWB range, resulting in distorted and inconsistent radiation patterns. Similarly, a large gain fluctuation of approximately 6 dBi is observed in printed elliptical monopole antenna with modified feeding structure for bandwidth enhancement [13], and a gain variation of almost 5 dBi is also observed in fork-shaped monopole [14], semi-elliptic monopole [15] antennas, UWB antennas with multiresonant split-ring loops [16] and multiple circular structures for wideband planar monopole antenna designs [17]. An eye-shaped antenna is proposed in [18] to achieve a bandwidth of 1.2–4.5 GHz but its gain is also reduced significantly from approximately 2 to 0 dBi within the operational band. As a result, a limited operating bandwidth regarding the radiation patterns and the peak gain is usually demonstrated.

Recently, antenna designers have focused on curve-based antenna geometries for bandwidth enhancement as various antenna shapes may be generated through the variations in parameters of antenna edge curve equation. In [19], a binomial curve function of order N has been proposed to design planar binomial curved monopole antenna for UWB applications. The patch shape has been varied from triangular ($N = 1$) to rectangular ($N = \infty$) by changing the order N of the binomial curve equation but binomial curved monopole antenna showed little influence over the antenna

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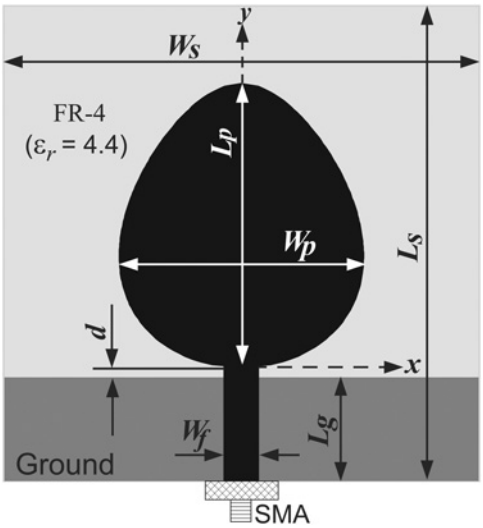


Fig. 1 *Geometry and parameters of proposed printed NECMA II configuration*

bandwidth with variation of order N. The crucial performance parameters of group delay and impulse response are also not discussed. In [20], using the same binomial curve as in [19], a CPW-fed wideband printed binomial curved slot antenna with maximum bandwidth of 99.7% (2.2–6.6 GHz) for rectangular ($N = \infty$) shape is proposed, but it did not cover whole UWB bandwidth (3.1–10.6 GHz). In [21], an elliptical patch is replaced with equivalent stepped edge polygon. It showed inconsistent radiation pattern over UWB range. Gain performance of the proposed antenna is also not discussed.

Moreover, the drawback of these planar antenna designs is that they are based on the lengthy trial and error method that involves computationally intensive full-wave electromagnetic simulations. In such circumstances, the designers are interested in having simple design formulas that provide a very good approximation to the final design when sophisticated electromagnetic (EM) analysis and design software packages are applied.

In this paper, a novel Newton's egg curve-based design of printed UWB monopole antenna is proposed. Section 2 describes the proposed design method. Simple design formulas are provided which are suitable for UWB antennas in the form of planar monopoles fed from a microstrip line. In Section 3, the simulation setup of proposed antenna based on inexpensive lossy FR-4 material is described. The Djordjević–Sarkar causal model [22] of frequency-dependent FR-4 material is used to ensure accurate simulation results. This method is useful for producing consistently causal and accurate models of broadband antennas. Several key parametric studies are performed for antenna optimisation and the bandwidth of various Newton's egg curved monopole antennas (NECMAs) has been discussed as the antenna design guidelines. An empirical formula is also proposed to approximately determine the frequency corresponding to the lower edge of $|S_{11}| \leq -10$ dB impedance bandwidth. To confirm the simulation results in Section 4, NECMAs with various structures are designed, fabricated and measured. Section 5 presents the conclusion.

2 UWB antenna configuration

The proposed microstrip line-fed printed NECMA is shown in Fig. 1. The different parametric variations in configurations of NECMAs are shown in Fig. 2. The proposed antenna is printed on double-sided FR-4 substrate ($W_s \times L_s = 40 \times 40$ mm^2) with a relative permittivity ε_r(4.4), loss tangent tan (δ) (0.018) and height h (1.6 mm). It is fed

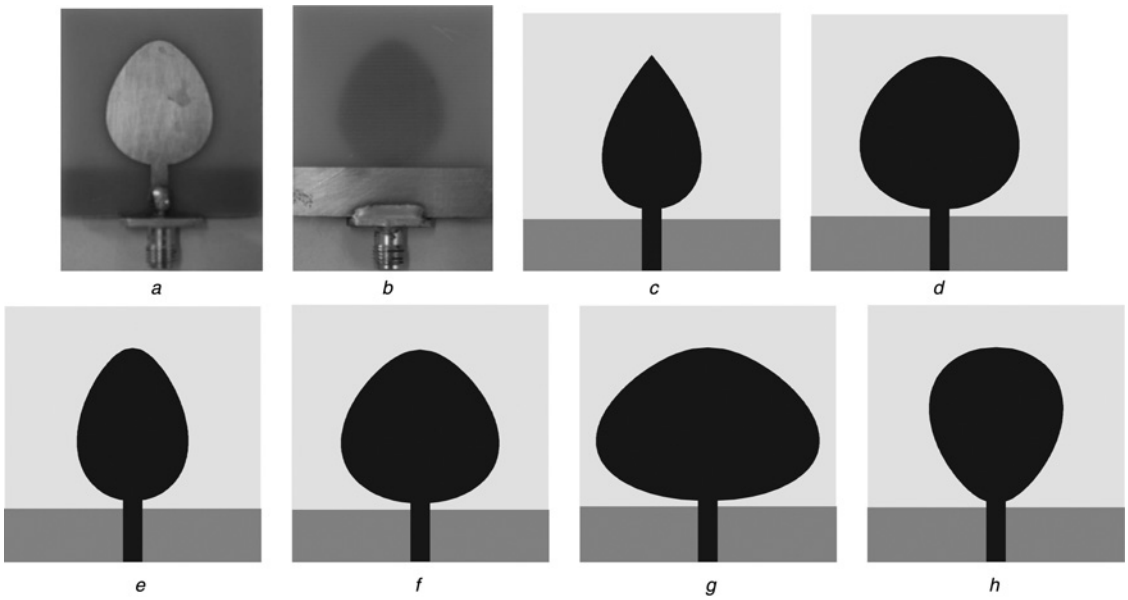


Fig. 2 *Various NECMA configurations for scaling factor s = 12*

a NECMA II (Top): $c = 0.35$, $b = 2$
b NECMA II (Back): $c = 0.35$, $b = 2$
c NECMA IV: $c = 0.35$, $b = 1$
d NECMA V: $c = 0.35$, $b = 3$
e NECMA VI: $c = 0.25$, $b = 2$
f NECMA VII: $c = 0.50$, $b = 2$
g NECMA VIII: $c = 1.0$, $b = 2$
h NECMA IX (Inverted egg): $c = 0.35$, $b = 2$

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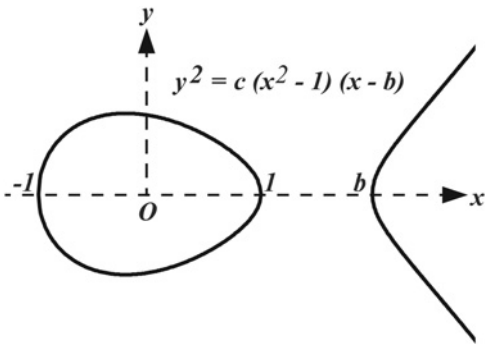


Fig. 3 *Generation of egg shape using Newton's diverging parabolas curve*

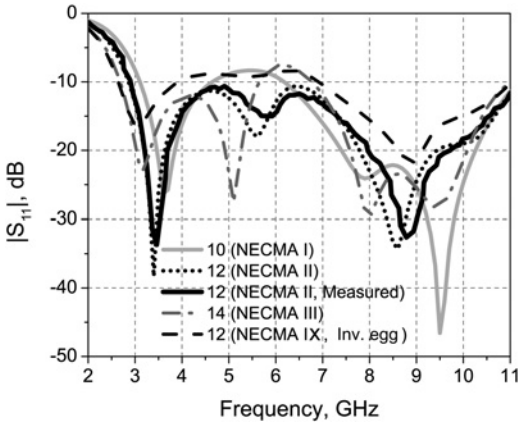


Fig. 4 *Simulated and measured reflection coefficients $|S_{11}|$ for various values of SF (s); (other parameters: $L_s \times W_s = 40 \times 40$ mm^2, $\varepsilon_r = 4.4$, $W_f = 3$ mm, $L_g = 9$ mm, $h = 1.6$ mm, $d = 0.15$ mm, $b = 2$, $c = 0.35$)*

by a 50 Ω microstrip line of width $W_f = 3$ mm which is excited by a 50 Ω SMA connector. The ground plane ($W_g \times L_g = 40 \times 9$ mm^2) is located on the back side of the substrate. The feed gap parameter d is the gap between the radiating patch and the ground plane which is optimised as $d = 0.15$ mm. The proposed fabricated NECMA II prototype is shown in Figs. 2*a* and *b*.

2.1 Design of egg curved patch

The egg-shaped patch geometry proposed in this paper is obtained from the Newton's diverging parabolas curve equation [23], shown in Fig. 3 as

$$y = \left[c(x^2 - 1)(x - b)\right]^{(1/2)}, \quad -1 \le x \le 1 \qquad (1)$$

Since (1) has the form of an egg, it is being referred here as Newton's egg curve. The Newton's egg curved patch obtained from (1) is rotated and moved along y-axis by feed length $L_f = L_g + d$ and then, scaled by a scaling factor (SF) of s in x-axis and y-axis directions. As shown in Fig. 2, different shapes of the radiating patch are formed for different sets of values for $c = 0.25$, 0.35, 0.50, 1.0 and $b = 1$, 2, 3.

3 Parametric study and BW comparison

The wide bandwidth of the NECMA results from the overlapping of the multiple resonances introduced by egg-shaped patch. The egg shape is related to parameters of the Newton's egg curve in (1), such as s, b and c along with feed gap parameter d, which mainly determine the antenna bandwidth. In this section, the frequency corresponding to the lower edge of −10 dB impedance bandwidth is determined first and then the antenna bandwidths for various values of s, b and c parameters are compared and discussed.

The simulation is carried out by using the 3-D full-wave electromagnetic simulation software Ansoft HFSS version 13.0 based on the finite-element method. The patch, ground plane and the microstrip feed line are the copper sheets of finite conductivity. The radiation box is kept at least a quarter wavelength (corresponding to the lowest frequency of operation of the proposed antenna) from the patch antenna in all directions. The lambda refinement is set as 0.25. The simulation frequency range is 2–11 GHz with step size of 0.1 GHz and the highest frequency of operation is selected as solution frequency. The antenna design and optimisation are achieved with tetrahedral meshes that typically include ∼50 000 tetrahedra after six mesh adaptation steps.

The lossy FR-4 dielectric substrate is used for fabrication of the proposed antenna. This low-cost composite material consists of glass fibres embedded in an epoxy resin [22]. For a successful modelling and design of the proposed antenna, it is important to consider the frequency variations of the dielectric parameters of FR-4 substrate because the frequency-dependent material properties can be critical at microwave/millimetre-wave frequencies. Therefore, the lossy FR-4 substrate is modelled using a Djordjević–Sarkar causal model in an Ansoft HFSS 13.0 3-D full-wave electromagnetic solver that has a broadband response which is faithful to measured results from DC to greater than 12 GHz [22]. This model satisfies the Kramers-Kronig conditions for causality, and so are preferred for applications where time-domain results are needed [22, 24]. The inputs to the Djordjević–Sarkar causal model are specified as the constant permittivity ($\epsilon_r = 4.4$) and loss tangent ($\tan\delta = 0.18$) of the dielectric material with measurement frequency as 1 GHz. The outputs from the Djordjević–Sarkar causal model are the expressions for permittivity and conductivity given in [22]. These expressions and zero loss tangent are used in place of the constant properties of the dielectric material [24]. It can be observed from Fig. 4 that measured reflection coefficient $|S_{11}|$ of lossy FR-4 material-based NECMA II prototype matches well with the simulated reflection coefficient $|S_{11}|$ of NECMA II design based on Djordjević–Sarkar causal model of frequency-dependent FR-4 material.

3.1 Scaling factor (s) and determination of lower edge frequency (for $|S_{11}| \le -10$ dB)

As discussed above, the egg shape obtained from Newton's egg curve (1) is scaled by a factor s in both x-axis and y-axis directions. Thus, it is obvious that the dimensions of designed Newton's egg curved patch will be determined in terms of scaling factor s. The proposed egg-shaped antenna structure is a variation of elliptical monopole antenna [25, 26]. Therefore, the lower cut-off frequency corresponding to $|S_{11}| \le -10$ dB for proposed NECMA is obtained from

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[26, 27] as follows

$$f_L = \frac{72}{\left(L_p + 0.125W_p + d\right)} \quad (2)$$

where L_p, W_p and d are in mm as indicated in Fig. 1 and f_L is in GHz. For $b = 2$ and $c = 0.35$, Newton's egg curve (1) may be rewritten as

$$y = \left[0.35(x^2 - 1)(x - 2)\right]^{(1/2)}, \quad -1 \le x \le 1 \quad (3)$$

Since Newton's egg curved patch is designed from (3) using scaling factor s, the dimensions L_p and W_p of Newton's egg curved patch are evaluated in terms of s from (3) as: $L_p = 2s$ and $W_p = 1.72s$. Thus, (2) may be expressed in terms of s as

$$f_L = \frac{72}{(2.215s + d)} \quad (4)$$

Since the simulated values of the lower cut-off frequency for $|S_{11}| \le -10$ dB are obtained for various values of s through full-wave EM simulation process as mentioned in Table 1, the more accurate expression for f_L of the form $p/(qs + d)$, where $d = 0.15$ and p and q are the unknown coefficients of the rational function, is obtained by using the curve fitting toolbox in MATLAB as

$$f_L' = \frac{0.9109}{(0.01425s + 0.15)} \quad (5)$$

The comparison between the calculated f'_L and the simulated lower cut-off frequency f'_L for $|S_{11}| \le -10$ dB corresponding to different values of scaling factor s is shown in Table 1. The simulated reflection coefficients $|S_{11}|$ of NECMAs (NECMA I, II, III) are plotted in Fig. 4, which indicate good match with the computed values of the lower cut-off frequency f'_L for $|S_{11}| \le -10$ dB in Table 1. The measured lower cut-off frequency of -10 dB operating bandwidth of NECMA II for $s = 12$ is 2.90 GHz (shown in Table 2), which matches well with corresponding simulated value of 2.85 GHz (mentioned in Table 1) as shown in Fig. 4. From Fig. 4, it is clear that the simulated lower cut-off frequency is reduced from 3.11 to 2.60 GHz (shown in Table 1) as scaling factor s is increased from 10 to 14, while other parameters are fixed. Fig. 4 also shows the plot of simulated $|S_{11}|$ for inverted egg shape configuration of NECMA IX with $s = 12$. Table 2 shows that the lowest cut-off frequency of 2.65 GHz for $|S_{11}| \le -10$ dB is obtained for NECMA IX compared to other NECMAs with $s = 12$, yet it did not cover the whole UWB bandwidth of 3.1–10.6 GHz. It is clear from Fig. 4 that NECMA IX has only two resonant modes compared to three resonant modes of NECMA II. Thus, the proposed NECMA II easily covers the entire UWB range with three resonant modes at 3.4, 5.5 and 8.5 GHz as indicated in Fig. 4. So, it is the overlapping of these resonances that leads to the UWB characteristic of proposed NECMA. Since the desired UWB bandwidth is covered with $s = 12$ for NECMA II (as shown in Table 2 and Fig. 4), s is selected as 12.

Table 1 Lower frequency edge ($|S_{11}| \le -10$ dB) for various SF (s); (other parameters: $L_s \times W_s = 40 \times 40$ mm^2, $\epsilon_r = 4.4$, $W_f = 3$ mm, $L_g = 9$ mm, $h = 1.6$ mm, $d = 0.15$ mm, $b = 2$, $c = 0.35$)

Antennas	SF(s)	Compu. f'_L, GHz	Simu. f'_L, GHz
NECMA I	10	3.11	3.11
NECMA II	12	2.84	2.85
NECMA III	14	2.61	2.60

Table 2 BW comparison for various NECMA configurations ($L_s \times W_s = 40 \times 40$ mm^2, $\epsilon_r = 4.4$, $d = 0.15$ mm, $W_f = 3$ mm, $h = 1.6$ mm, $L_g = 9$ mm)

Antennas	Parameters			%BW ($|S_{11}| \le -10$ dB), frequency range, GHz	
	s	b	c	Simulated	Measured
NECMA I	10	2.0	0.35	39.07%, 3.11–4.62; 56.56%, 6.15–11.0	—
NECMA II	12	2.0	0.35	117.69%, 2.85–11.0	116.55%, 2.90–11.0
NECMA III	14	2.0	0.35	73.79%, 2.60–5.64; 47.61%, 6.77–11.0	—
NECMA IV	12	1.0	0.35	36.13%, 2.88–4.15; 63.79%, 5.68–11.0	—
NECMA V	12	3.0	0.35	68.18%, 2.90–5.90; 48.59%, 6.70–11.0	—
NECMA VI	12	2.0	0.25	40.23%, 2.78–4.18; 71.27%, 5.22–11.0	—
NECMA VII	12	2.0	0.50	68.16%, 2.95–6.0; 49.29%, 6.65–11.0	—
NECMA VIII	12	2.0	1.00	37.58%, 3.35–4.90; 41.64%, 6.75–10.30	—
NECMA IX (Inv. egg)	12	2.0	0.35	39.39%, 2.65–3.95; 45.13%, 6.95–11.0	—

3.2 Newton's egg curve parameters b and c

The antenna characteristics are defined by the excited operating modes which depend mainly on the shape and dimensions of the antenna structure. If two or more adjacent modes are well excited simultaneously, the bandwidth can be twice or more than that for the single resonance. Hence, the curve of a printed antenna edge is a key parameter which affects the antenna characteristics. Thus, suitable selection of proposed Newton's egg curve parameters b and c enhances the impedance bandwidth to UWB range. The simulated results of reflection coefficient for the proposed egg curved monopole antennas, with different values of Newton's egg curve parameter c, are shown in Fig. 5, while keeping other parameters fixed. The corresponding values of antenna bandwidth for $|S_{11}| \le -10$ dB are also listed in Table 2. It is clear that the lower edge frequency of -10 dB operating bandwidth of NECMAs is reduced from 3.35 to 2.78 GHz as c is decreased from 1 to 0.25 but whole UWB range is covered for $c = 0.35$. So, $c = 0.35$ is selected as optimum value for UWB applications. Similarly, the simulated results of reflection coefficient for proposed NECMAs with different values of Newton's egg curve parameter b are shown in Fig. 6, while keeping other parameters fixed. When the value of b is varied from 1 to 3, then the optimum value of b for UWB range is obtained as 2 as shown in Table 2.

3.3 Feed gap parameter d

The gap between the radiating patch and the ground plane affects the impedance bandwidth as it acts as a matching network. The impedance bandwidth of the proposed

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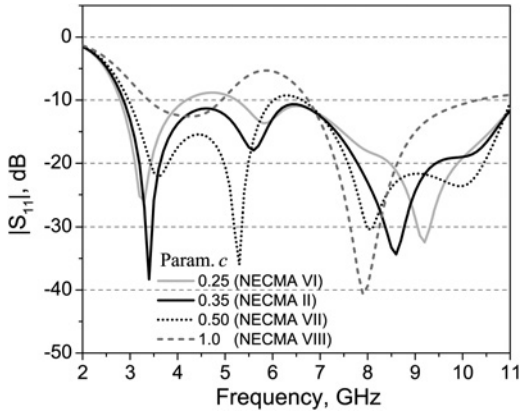


Fig. 5 *Simulated reflection coefficients $|S_{11}|$ for various values of Newton's egg curve parameter c; (other parameters: $L_s \times W_s = 40 \times 40$ mm^2, $\epsilon_r = 4.4$, $W_f = 3$ mm, $L_g = 9$ mm, $h = 1.6$ mm, $d = 0.15$ mm, $b = 2$, $s = 12$)*

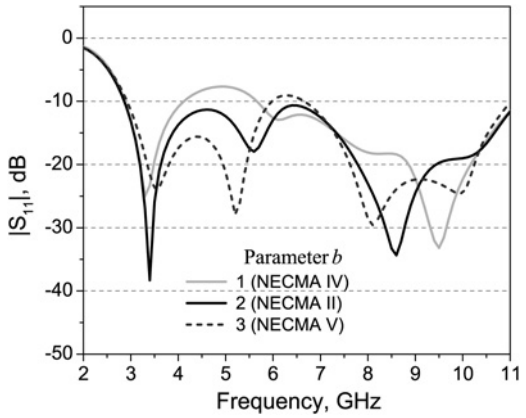


Fig. 6 *Simulated reflection coefficients $|S_{11}|$ for various values of Newton's egg curve parameter b; (other parameters: $L_s \times W_s = 40 \times 40$ mm^2, $\epsilon_r = 4.4$, $W_f = 3$ mm, $L_g = 9$ mm, $h = 1.6$ mm, $d = 0.15$ mm, $c = 0.35$, $s = 12$)*

antenna at different gaps is shown in Fig. 7. It can be observed that the lower resonant frequency of −10 dB impedance bandwidth is independent of the feed gap but the upper resonant frequency is dependent on it. The optimum impedance matching is obtained with $d = 0.15$ mm. At $d = 0.15$ mm, the capacitance that results from the spacing between the edge of the ground plane and radiating patch reasonably balances the inductance of the antenna.

3.4 Surface current distribution

The simulated surface current distributions for the proposed NECMA II configuration at the frequencies 3.4, 5.5 and 8.5 GHz are shown in Fig. 8. In monopole antennas, both the monopole radiator and the ground plane contribute to the radiating fields. At 3.4 GHz, Fig. 8*a* shows that the current is primarily distributed over the monopole with a small horizontal component of current induced at the edge of the ground plane near the radiating patch, which is similar to the current distribution of a printed resonant monopole antenna. At 5.5 GHz, it is clear from Fig. 8*b* that the currents are mainly concentrated near the upper edge of the ground plane, while on top of the structure, currents are

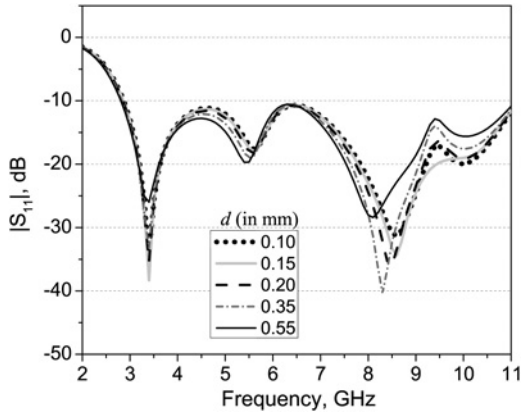


Fig. 7 *Simulated reflection coefficients $|S_{11}|$ for various values of feed gap parameter d; (other parameters: $L_s \times W_s = 40 \times 40$ mm^2, $\epsilon_r = 4.4$, $W_f = 3$ mm, $L_g = 9$ mm, $h = 1.6$ mm, $b = 2$, $c = 0.35$, $s = 12$)*

primarily distributed along the periphery of the edge of the NECMA II patch and feed line. At 8.5 GHz, it can be observed from Fig. 8*c* that the current is mainly distributed around the gap between the monopole and the ground. The radiating slots can be thought to form between the lower edge of the patch and the ground plane [28, 29]. Therefore, it behaves more like a slot antenna at 8.5 GHz. It is also observed from Fig. 7 that the effect of feed gap width d is more dominant at higher frequencies compared to the lower frequencies.

4 Results and discussion

The proposed NECMAs were initially simulated and then, fabricated on a low-cost FR-4 substrate of thickness $h = 1.6$ mm and relative permittivity $\epsilon_r = 4.4$. The other antenna design parameters are listed in Table 2. The $|S_{11}|$, radiation patterns and gain of the proposed NECMA II shown in Fig. 4, Figs. 9 and 10, respectively, have been measured using the antenna measurement system, Satimo Starlab microwave anechoic chamber. When an antenna is placed in the Starlab for measurement, a coaxial feeding cable is always required to connect the antenna to the system. In the measurements of small monopole antennas, the electrical size of the antenna ground plane becomes relatively small at low frequencies compared to the wavelength. So, the currents flow back from the antenna to the feeding cable, giving rise to 'secondary radiation' [30] and causing inaccuracy and ripples in $|S_{11}|$ measurement. To reduce the effects on the measured $|S_{11}|$, the feeding cable in the Starlab system is covered with EM suppressant tubing to absorb the currents flowing back to the cable and radiation from the cable. As a result, the antenna is well matched across the desired frequency range and a balun is not needed to reduce the return loss. It is observed from Fig. 4 that the proposed antenna NECMA II has achieved measured impedance bandwidth of 2.9–11.0 GHz for UWB applications. In this section, experimental and simulation results are presented.

4.1 Radiation patterns

The measured far-field *E*-plane (or *yz*-plane) and *H*-plane (or *xz*-plane) radiation patterns of the proposed antenna NECMA II are shown at 3.5, 5.5 and 9.0 GHz as shown in Fig. 9. The copolarised components in the *E*-plane (or *yz*-plane)

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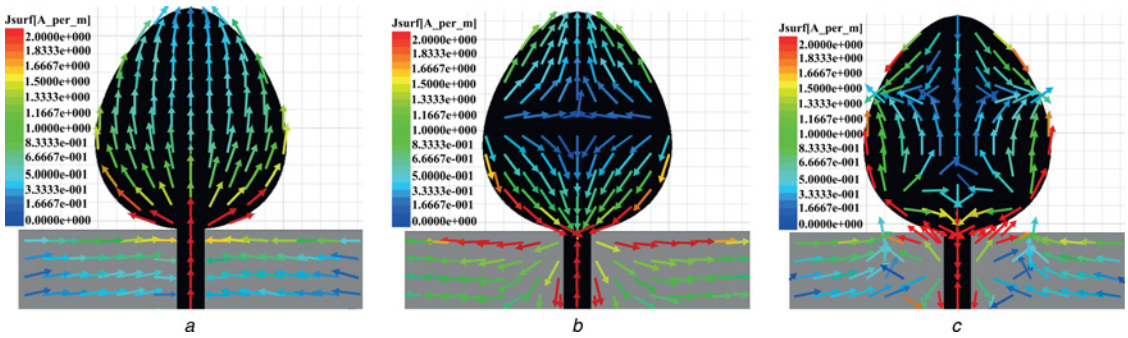


Fig. 8 *Simulated current distributions for the proposed NECMA II*

a 3.4 GHz
b 5.5 GHz
c 8.5 GHz

and *H*-plane (or *xz*-plane) are E_θ and E_ϕ, respectively. It can be observed that the proposed NECMA II shows symmetrical and stable omnidirectional radiation patterns in *H*-plane at all frequencies in the entire UWB range, whereas bidirectional patterns in the *E*-plane. Fig. 9*a* shows that the copolarised *E*-plane radiation pattern is close to a typical figure-of-eight at lower frequency 3.5 GHz which is similar to a conventional dipole because of a small ground plane [31]. At higher frequencies, the copolarised *E*-plane pattern is slightly rotated. Fig. 9*a* also shows that the crosspolar radiation is very low at lower frequency 3.5 GHz and the value of crosspolarisation goes up with an increase in frequency as shown in Figs. 9*b* and *c*, which is a result of the increasing horizontal component of the surface currents near the upper edge of the ground plane as also indicated in Figs. 8*b* and *c*. However, it is obvious from Fig. 8 that the horizontal component of the current is induced only at the upper edge of the ground plane near the radiating patch at all frequencies. Therefore, lower crosspolar radiation (<−20 dB) is observed in all directions

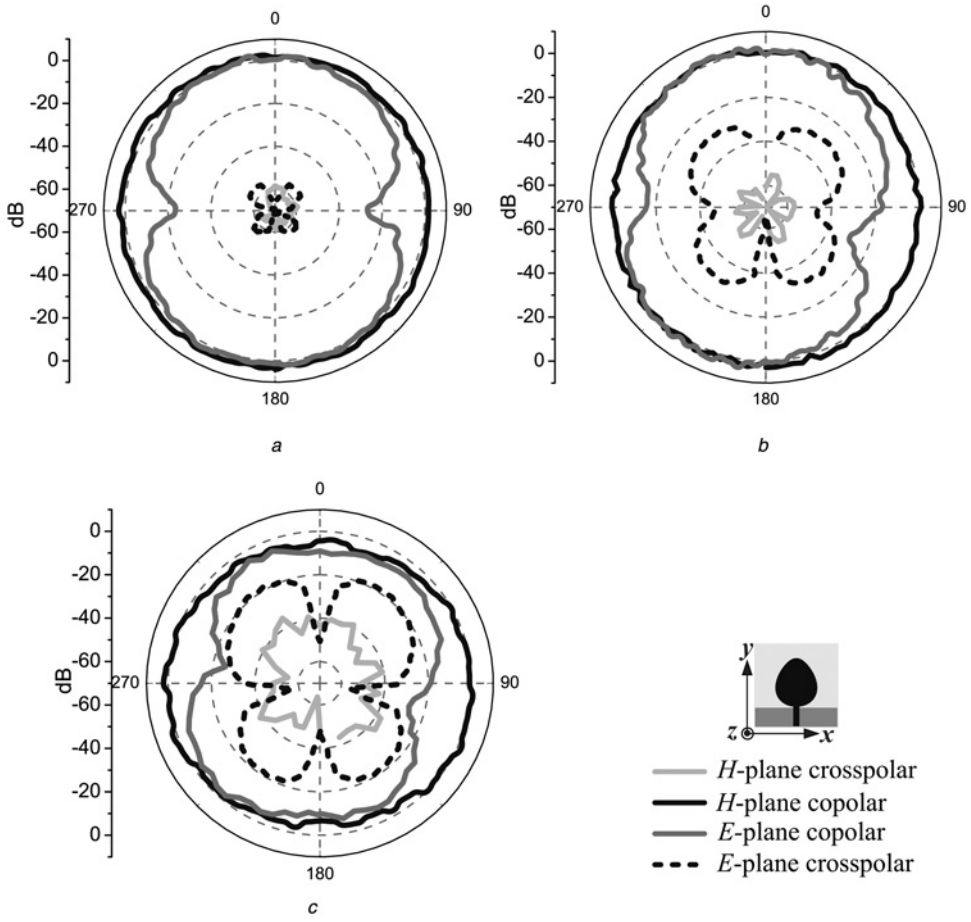


Fig. 9 *Measured radiation patterns for the proposed NECMA II*

a 3.5 GHz
b 5.5 GHz
c 9.0 GHz

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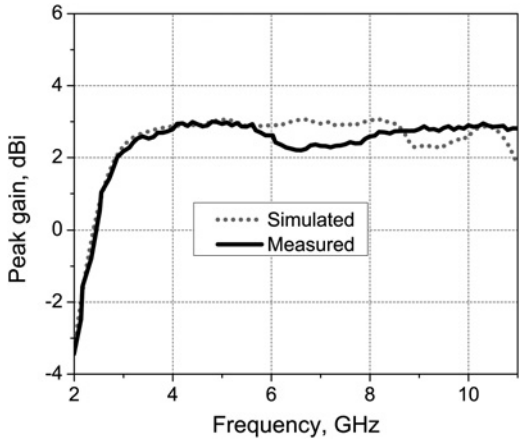


Fig. 10 *Comparison of simulated and measured gains of proposed NECMA II*

over the entire UWB bandwidth compared to crosspolar radiation level of printed elliptical monopole antenna with modified feeding structure in [13]. Hence, the radiation patterns of the proposed antenna are suitable for various UWB applications.

4.2 Gain

A good match between the measured and simulated gains of the proposed NECMA II across the entire UWB frequency band is shown in Fig. 10. The comparison between simulated and measured gains of the proposed NECMA II with planar binomial curved monopole antenna [19] is also shown in Table 3. It is clear that the proposed NECMA II provides larger and more consistent gain of 2.2–2.9 dBi measured over whole UWB bandwidth as compared to planar binomial curved monopole antenna ($N = 4$) [19]. The proposed NECMA II also showed lesser gain variation of 0.7 dBi over whole UWB band as compared to approximate 5 dBi gain variation of printed CPW-fed semi-elliptic monopole [15], approximate 6 dBi gain variation of elliptical monopole with modified feed arrangement [13] and 1.3 dBi gain variation of printed CPW-fed binomial curved slot antenna ($N = 6$) [20] whereas the gain performance of elliptical dipole with stepped edge polygon is not discussed in [21]. The gain stability of the proposed NECMA II also indicates the consistency of radiation patterns over the entire UWB bandwidth.

Table 3 Performance comparison of various printed curved antennas

Printed curved antenna config.	Patch dimensions ($L_s \times W_s$) mm^2	Measured gain, dBi, frequency range, GHz
planar NECMA II	40 × 40	2.2–2.9, 2.9–11.0
printed CPW-fed semi-elliptic monopole [15]	34 × 25	0.5–5.5, 3.0–11.0
planar binomial curved monopole ($N = 4$) [19]	70 × 46	–0.9–1.5, 3.0–11.0
printed binomial curved slot antenna ($N = 6$) [20]	35 × 40	3.5–4.8, 2.45–5.29

4.3 Time-domain results

UWB systems must ensure the undistorted reception of the transmitted signal. So, the measurement of transmission coefficient S_{21} is performed using an Agilent E5071C ENA network analyser in the anechoic chamber by exciting two identical prototypes of the proposed NECMA II kept in the far field at a distance $R = 60$ cm ($\sim 6\lambda$ at the lowest operating frequency), for face-to-face and side-by-side orientations. The group delay τ_g is defined as the negative derivative of the antenna phase angle with respect to frequency [28], $\tau_g(\omega) = -\partial\phi/\partial\omega = -\partial\phi/2\pi\partial f$. Small variations in group delay, defining a flat response or linear phase within a particular frequency range, suggest that waveform distortions in the time domain of transmitted or received pulses will be small [28]. The measured group delay of proposed NECMA II remains stable in both face-to-face and side-by-side orientations with variation less than 0.5 ns over whole UWB band, as shown in Fig. 11. This means that the phase centre of the antenna remains very close to the antenna plane over the entire UWB bandwidth. The antenna transfer function is expressed as

$$H(\omega) = \sqrt{\frac{2\pi RcS_{21}(\omega)e^{j\omega R/c}}{j\omega}} \tag{6}$$

where c is the free space velocity and R is the distance between the two antennas [32] and is plotted in Fig. 12. It

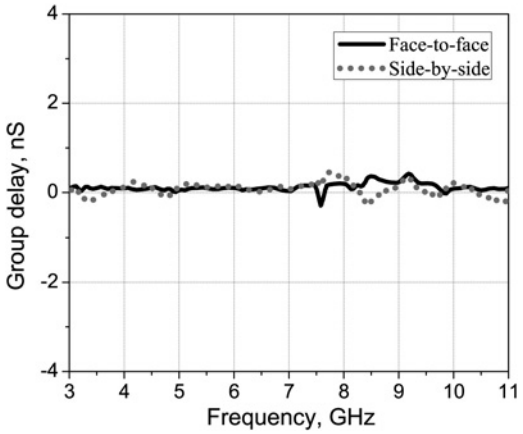


Fig. 11 *Measured group delay of proposed NECMA II prototypes*

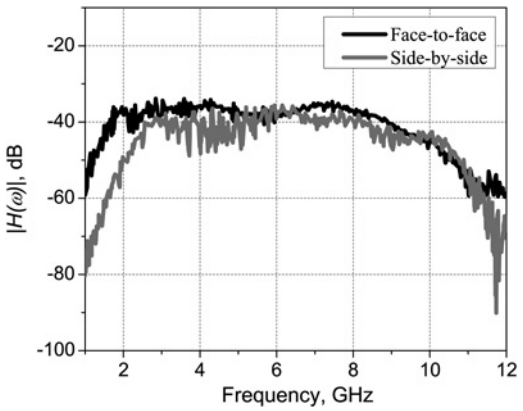


Fig. 12 *Measured transmission characteristics of proposed NECMA II prototypes*

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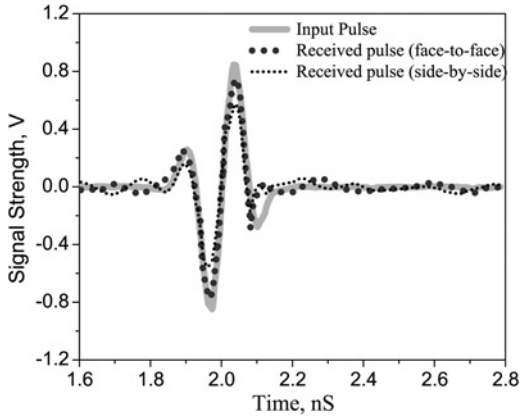


Fig. 13 *Input pulse and received pulse in time domain for a UWB link with two identical NECMA II prototypes*

shows fairly flat magnitude throughout the UWB band for both the orientations. Transient response of the antenna is studied by convoluting the modulated Gaussian monocycle pulse defined as

$$s(t) = \sin(2\pi f_c t)\mathrm{e}^{-(t/\sigma)^2} \tag{7}$$

with $h(t)$, the inverse Fourier transform of $H(\omega)$ in (6). Spectrum of source pulse in (7) fully covers the FCC band and comply with the emission standards specified with carrier frequency $f_c = 6.85$ GHz and pulse duration parameter $\sigma = 91$ ps. This mask limits power emission of the UWB radios to avoid possible interference with other radio systems operating within the same band [1]. Thus, the received pulse is computed as

$$r(t) = s(t) \otimes h(t) \tag{8}$$

The input and received waveforms for the face-to-face and side-by-side orientations of the antenna are shown in Fig. 13. Although the maximum magnitude of the waveform for the side-by-side orientation is less than the face-to-face orientation, both the waveforms retain the information contained in the transmitted signal with minimum dispersion. These results indicate that the antenna has good linear transmission performances and is useful for multiband UWB or impulse UWB radio applications.

5 Conclusion

Printed NECMA of size 40×40 mm^2 is proposed, designed and fabricated on low cost FR-4 substrate. The lossy FR-4 substrate is modelled using Djordjević–Sarkar causal model. The numerical simulations have been validated experimentally with good agreement. Simple design formulas are provided for UWB antenna design. The bandwidth of the proposed NECMA II is measured as 116.55% (2.9–11.0 GHz) for $|S_{11}| \leq -10$ dB. It is observed that the proposed antenna has improved gain and more consistent radiation patterns compared to planar binomial curved monopole antenna. Moreover, with different parametric variations in proposed Newton's egg curve equation, NECMAs with different bandwidths are also obtained for various wideband applications. The proposed NECMA II provides stable and almost omnidirectional radiation patterns in *H*-plane, along with stable group delay and nearly flat gain (2.2–2.9 dBi) over the entire UWB bandwidth (3.1–10.6 GHz) which makes it a suitable candidate for UWB applications.

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